

INTRODUCTION

Rivers are among the most biodiverse ecosystems on the planet. A river is a natural stream of water that flows from a high source to another body of water and are crucial to human survival by providing potable water to people globally. In the UK, this is evident as roughly 66% of drinking water stems from rivers, lakes and reservoirs.

Lakes and rivers require low levels of nutrients (such as NO_3^- and PO_4^{3-}) and pollutants to support wildlife, so we have a role to play in preventing the bioaccumulation of these substances - both to provide potable water worldwide and to allow rivers to remain a viable habitat for a wide range of animals and organisms.

Rivers also carry foul water away from our homes, moving sewage via combined sewer overflow systems and helping prevent sewage backup during heavy rainfall, which keeps our streets clean. These points motivate our study, in which we investigated the upstream, downstream and discharge concentrations of algae, bacteria, NO_3^- and PO_4^{3-} . High concentrations of these indicate low water quality, conflicting with the societal interests outlined above.

METHODS

Water samples were collected from three locations along the river: upstream, downstream and at the point of discharge.

1. Algal count (microscopy)

- Concentrate the algae sample
- Quantify algae using a haemocytometer
- Multiply the count by 1000 to determine total algae in the original sample

2. Nutrient concentration (spectrophotometry)

- Prepare upstream, downstream and discharge samples
- Measure absorbance of all samples at 840 nm
- Construct and read from a calibration curve to determine phosphate and nitrate concentration

3. Bacterial count (spread plates)

- Prepare serial dilutions of the three samples
- Spread diluted samples on agar plates, and E. coli samples on EMB agar plates
- Count bacterial colonies to calculate bacteria count in the original sample

4. Antibiotic resistance (PCR & gel electrophoresis)

- Set up PCR to screen for antibiotic-resistance genes
- Run gel electrophoresis on an agarose gel
- Photograph and analyse the gel

RESULTS

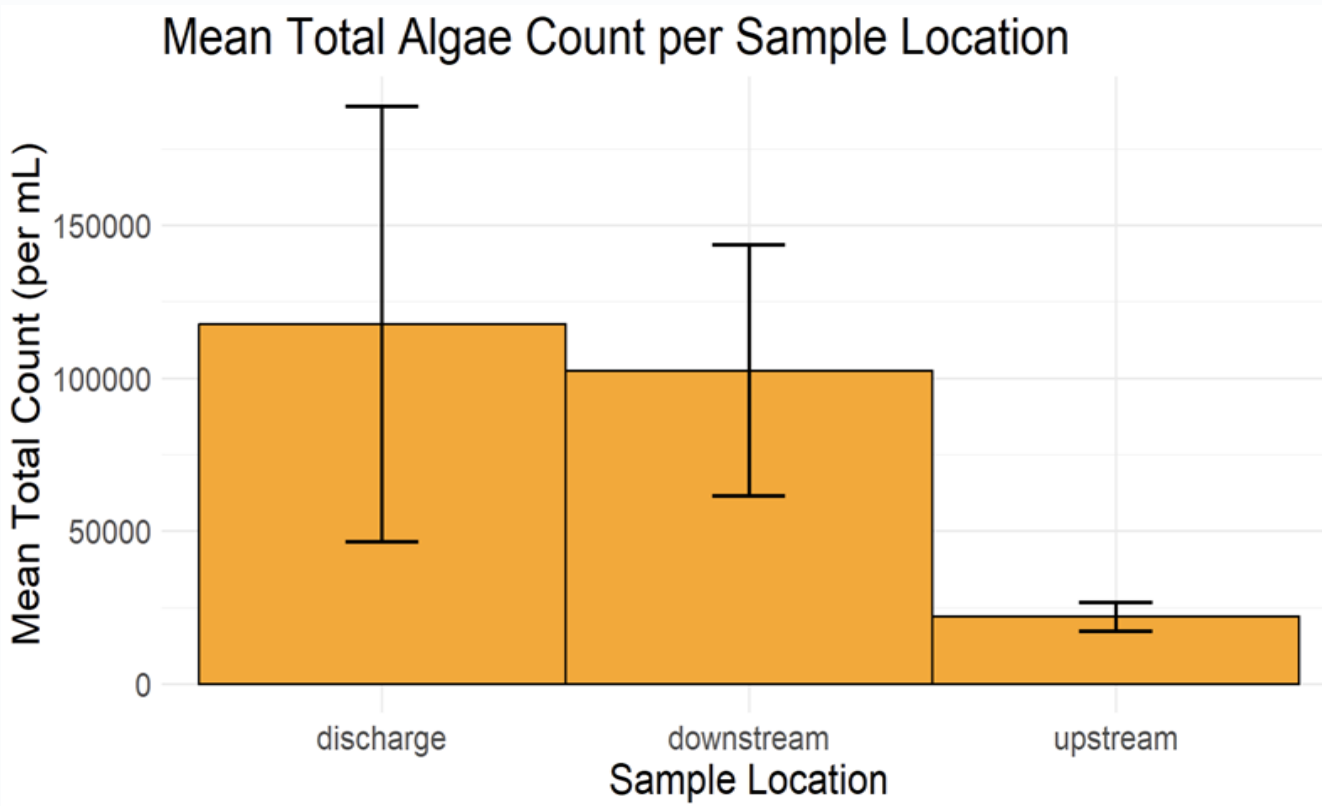


Figure 1. Mean total algal count per sample location, determined by microscopy and haemocytometer counts. Error bars show $\pm$ standard deviation.

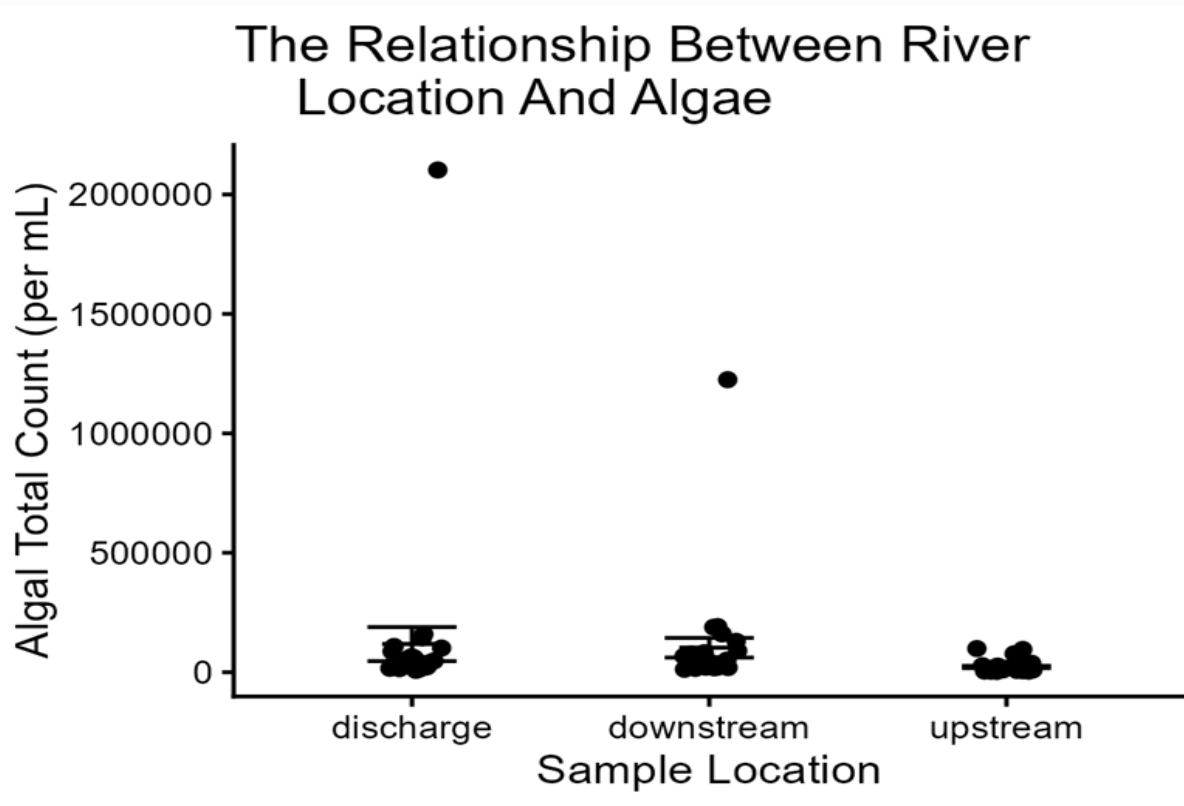


Figure 2. Relationship between river location and total algal count per sample.

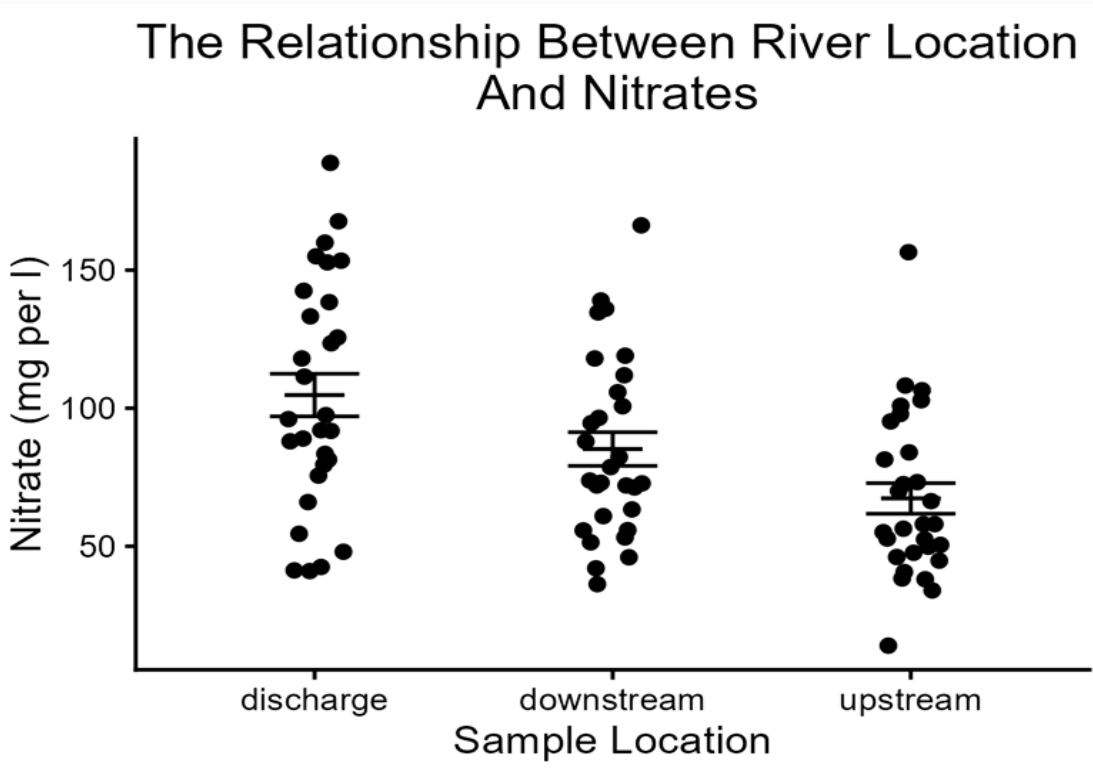


Figure 3. Relationship between river location and nitrate concentration (mg/l).

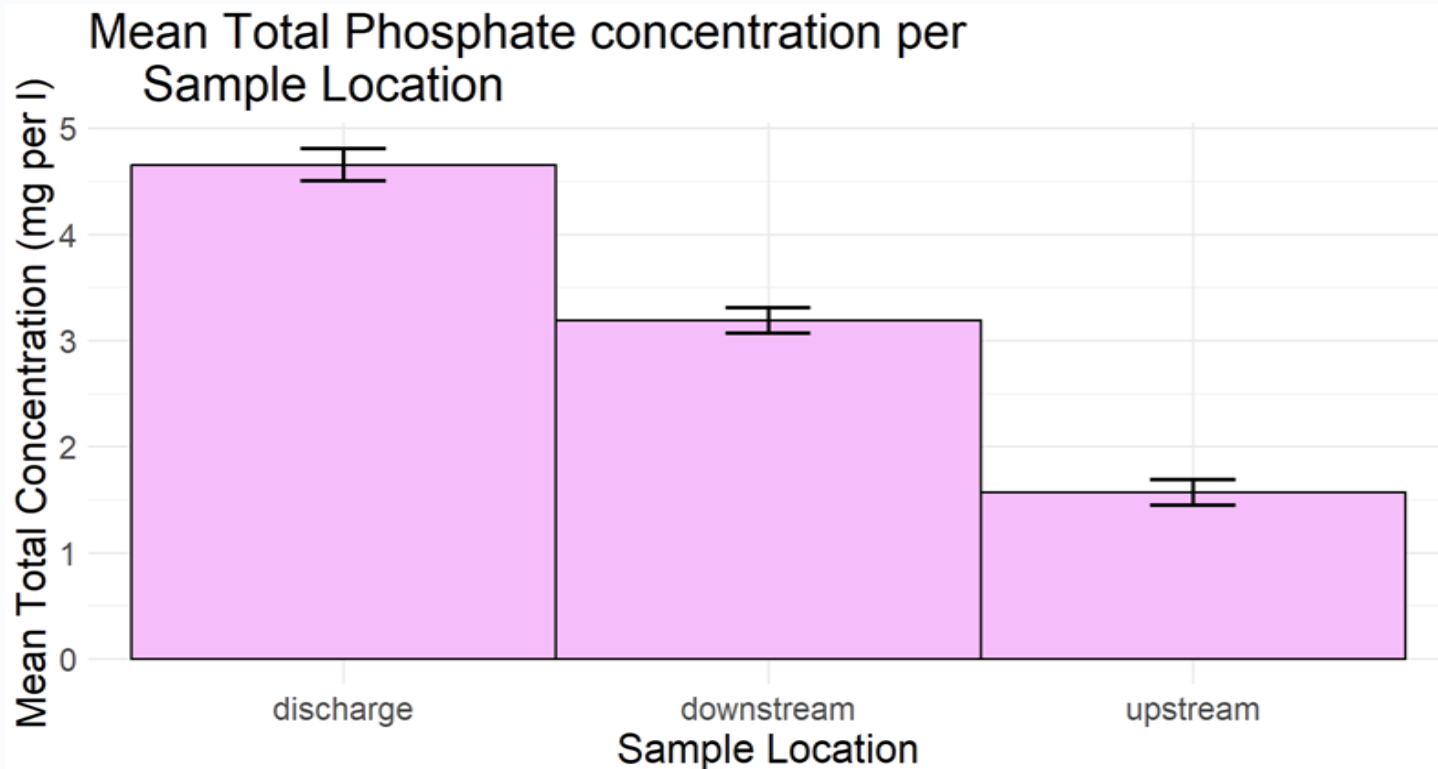


Figure 4. Mean total phosphate concentration (mg/l) per sample location. Error bars show $\pm$ standard deviation.

DISCUSSION

The results demonstrate spatial variation in river water quality, with the discharge site showing the highest nutrient concentrations and elevated algal abundance.

Elevated nitrate and phosphate alongside increased algal abundance are consistent with nutrient enrichment and a potential risk of eutrophication. Subsequent decomposition of algal biomass may increase oxygen demand and reduce dissolved oxygen, potentially affecting aquatic biodiversity. However, algal counts alone cannot confirm an algal bloom or the presence of toxin-producing cyanobacteria, and manual counting introduces potential observer-dependent variation.

Elevated nutrient concentrations at the discharge and downstream sites suggest increased nutrient loading, although the specific source cannot be established from these data alone.

Nutrient enrichment may originate from sewage, agricultural runoff or other anthropogenic inputs. Antibiotic-resistant bacteria were not detected under the experimental conditions used; however, this does not demonstrate their complete absence because detection depends on the organisms and resistance markers targeted.

Overall, the findings highlight spatial differences in river quality and support the value of continued environmental monitoring. Future work should incorporate larger and seasonal sample sets, dissolved oxygen measurements, statistical testing and identification of potentially harmful cyanobacteria.

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